ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 7

NATURAL SCIENCES

PRACTICE EXAM 3 · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central Grade 7 Natural Sciences practice exam 3.

Written by Academic Central for practice, following the CAPS curriculum for Grade 7. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Accept answers in the learner's own words if the science is correct. Accept other correct examples.

SECTION A

QUESTION 1

1 1.1 D ✓ 1.2 B ✓ 1.3 C ✓ 1.4 A ✓ 1.5 B ✓ 1.6 D ✓ 1.7 A ✓ 1.8 C ✓ 1.9 C ✓ 1.10 A ✓ (10)

QUESTION 2

2.1 False ✓ – moist (smooth) ✓ (2)

2.2 True ✓✓ (2)

2.3 False ✓ – radiation ✓ (2)

2.4 True ✓✓ (2)

2.5 False ✓ – reflects light from the Sun ✓ (2)

TOTAL SECTION A: [20]

SECTION B

QUESTION 3

3.1 Any TWO: water ✓; air (oxygen or carbon dioxide) ✓; energy (sunlight or food); soil / nutrients; a suitable temperature; shelter (2)

3.2 (a) Mammal ✓ (b) Reptile ✓ (c) Amphibian ✓ (d) Bird ✓ (4)

3.3 Any TWO: molluscs (snails, octopuses) ✓; worms ✓; cnidarians (jellyfish, corals); echinoderms (starfish); sponges (2)

3.4 (a) Any TWO: small, dull petals or none ✓; no scent or nectar; large, feathery stigmas hanging outside the flower ✓; anthers hanging outside; large amounts of light, smooth pollen (3)
(b) Most of the pollen is blown away and never lands on a stigma of the same kind ✓

3.5 (a) A seed ✓ (b) A fruit ✓ (2)

3.6 Girls, any TWO: breasts develop ✓; hips widen ✓; menstruation starts (4)
Boys, any TWO: voice deepens (breaks) ✓; facial hair grows ✓; shoulders broaden; testes start producing sperm

3.7 Harmful substances (nicotine, alcohol) pass from the mother's blood through the placenta to the baby ✓ and can harm its growth and development, e.g. low birth mass or brain damage ✓. (2)

3.8 A group of organisms that are similar and can breed with each other to produce fertile offspring ✓ (1)

[20]

QUESTION 4

4.1.1 Filtration ✓ (1)

4.1.2 A filter paper ✓ B funnel ✓ D residue (mud / sand) ✓ (3)

4.2 (a) Mixture ✓ (b) Pure substance ✓ (c) Mixture ✓ (d) Pure substance ✓ (4)

4.3 (a) P ✓ (b) Q ✓ (c) R ✓ (3)

4.4 (a) In order of increasing atomic number (the number of protons) ✓ (4)
(b) Any TWO: oxygen ✓; nitrogen ✓; hydrogen; helium; neon; chlorine; argon
(c) He ✓

4.5 (a) Any TWO: they do not rot (not biodegradable) and fill landfill sites ✓; animals eat them or get caught in them and die ✓; they block drains and pollute rivers and oceans; burning them releases poisonous gases (3)
(b) Any ONE: use cloth or reusable shopping bags; reuse plastic bags; recycle them ✓

4.6 (a) Sieving (or filtration) ✓ (b) Evaporation ✓ (2)

[20]

QUESTION 5

5.1 (a) Conduction ✓ (b) Convection ✓ (c) Radiation ✓ (3)

5.2 Copper is a very good conductor of heat, so heat spreads quickly and evenly to the food ✓ (1)

5.3 (a) Black surfaces absorb radiant heat from the Sun best ✓✓ (3)
(b) To reduce heat loss so the water stays hot ✓

5.4 The child has the most potential energy at the highest points of the swing ✓ and the most kinetic energy at the lowest point ✓. Between these points potential energy changes into kinetic energy and back again (some is lost as heat and sound) ✓. (3)

5.5 (a) 2 000 − 1 800 = 200 J ✓ (b) It heats the kettle itself and the surrounding air (and some becomes sound) ✓ (2)

5.6 (a) Non-renewable ✓ (b) Renewable ✓ (c) Non-renewable ✓ (3)

5.7 Power station → step-up transformer → transmission lines → step-down transformer → homes ✓✓✓ (one mark for every correct pair in sequence, maximum 3) (3)

5.8 (a) Eskom ✓ (b) Any ONE: nuclear (Koeberg) ✓; hydro-electric; pumped storage; gas turbines; wind (2)

[20]

QUESTION 6

6.1.1 (a) P ✓ (b) R ✓ (2)

6.1.2 At R the Earth is between the Sun and the Moon ✓, so the whole sunlit half of the Moon faces the Earth ✓. (2)

6.2 In December the Southern Hemisphere is tilted towards the Sun ✓, so the Sun's rays strike it more directly (concentrated on a smaller area) and the days are longer ✓. (2)

6.3 21 or 22 December ✓ (1)

6.4 (a) Two ✓ (b) The Moon is much closer to the Earth than the Sun ✓ (2)

6.5 The Sun's rays can permanently damage your eyes (cause blindness) ✓ (1)

6.6 (a) The Earth is at the centre of the universe (geocentric model) ✓ (4)
(b) The Sun is at the centre and the planets move around it (heliocentric model) ✓
(c) Used a telescope to see the moons of Jupiter / the phases of Venus / craters on the Moon ✓
(d) Explained that gravity keeps the planets in orbit around the Sun (and the Moon around the Earth) ✓

6.7 South ✓ (1)

6.8 (a) Radio waves ✓ (3)
(b) Any TWO: few people, so very little radio interference from cell phones and radios ✓; dry, clear conditions ✓; a good view of the centre of our galaxy from the Southern Hemisphere; flat, open land

6.9 The Southern Hemisphere is tilted away from the Sun ✓, so the Sun rises later, sets earlier and moves along a shorter, lower path across the sky ✓. (2)

[20]

TOTAL SECTION B: [80]

TOTAL: 100

END OF MARKING GUIDELINE